

Engineering Molecular Assembly for High Performance Plastic Thermoelectrics

Dongyang Wang, Daoben Zhu, and Chong-an Di*



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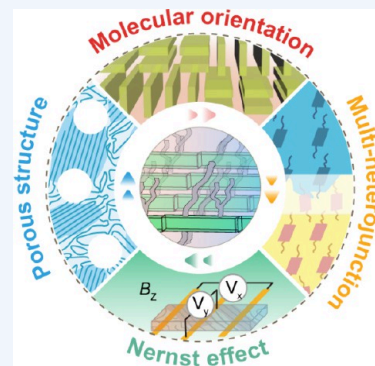
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CONSPECTUS: The escalating global energy crisis, coupled with the environmental impact of conventional energy consumption, has intensified the pursuit of green and sustainable energy solutions. Converting low-grade heat into electricity using flexible, lightweight, and solution-processable polymeric thermoelectrics offers unique opportunities for next-generation wearable and portable power systems. Early studies in this field predominantly emphasized molecular design, optimizing conjugated backbones and side chains to enhance the charge transport and Seebeck coefficients. These efforts yielded valuable insights into the relationships among molecular structure, electronic states, and thermoelectric performance. More recently, molecular assembly engineering has attracted growing interest driven by recognition of how microstructural order and hierarchical morphology affect carrier mobility and energy filtering. Strategies such as controlled self-assembly, directional alignment, and interface engineering have proven highly effective, enabling property enhancement beyond the limits of molecular design alone. This shift has not only produced high-performance polymeric thermoelectric materials but also broadened their functional scope, opening opportunities for integration into flexible and versatile energy systems.

The “phonon-glass, electron-crystal” (PGEC) concept envisions an ideal assembly that combines the intrinsically low thermal conductivity of amorphous glasses with the exceptional charge transport of crystalline solids. Guided by these design principles, our recent work has addressed the coupled optimization of charge and thermal transport while exploring novel functional capabilities in polymeric thermoelectrics. To mitigate the persistent trade-off between the Seebeck coefficient (S) and electrical conductivity (σ), we developed a mixed-orientation strategy in which bimodal molecular orientation generates interfacial weak hydrogen bonds, promoting efficient chemical doping, improved molecular ordering, and increased density of states. This synergistic effect yielded simultaneous enhancements in S and σ , achieving a peak figure of merit (ZT) more than four times higher than that of single-orientation films. In addition, we tackled the underestimated role of thermal conductivity (κ) by introducing a heterogeneous assembly approach for high-mobility polymers. Incorporating porous architectures induced localized vibrational scattering, lowering lattice κ and raising ZT to 0.52. Building further, we introduced a polymeric multi-heterojunction (PMHJ) architecture via cross-linking-assisted assembly, where pronounced size effects and interfacial diffuse scattering reduced κ to $0.10 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, delivering a record-high ZT of 1.28, comparable to that of commercial Bi_2Te_3 materials in the near-room-temperature region. Extending beyond performance metrics, we revealed anomalously large Nernst coefficients in doped polymers, 2–3 orders of magnitude above classical Fermi-liquid predictions—paving the way for advanced lateral organic thermoelectric devices.



KEY REFERENCES

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- Wang, D. Y.; Ding, J. M.; Dai, X. J.; Xiang, L. Y.; Ye, D. K.; He, Z. H.; Zhang, F. J.; Jung, S. H.; Lee, J. K.; Di, C. A.; et al. Triggering ZT to 0.40 by Engineering

Orientation in One Polymeric Semiconductor. *Adv. Mater.* **2023**, 35, e2208215.² A bimodal molecular-orientation regulation strategy can break the intrinsic trade-off between the Seebeck coefficient and electrical conductivity, thereby enhancing the thermoelectric figure of merit.

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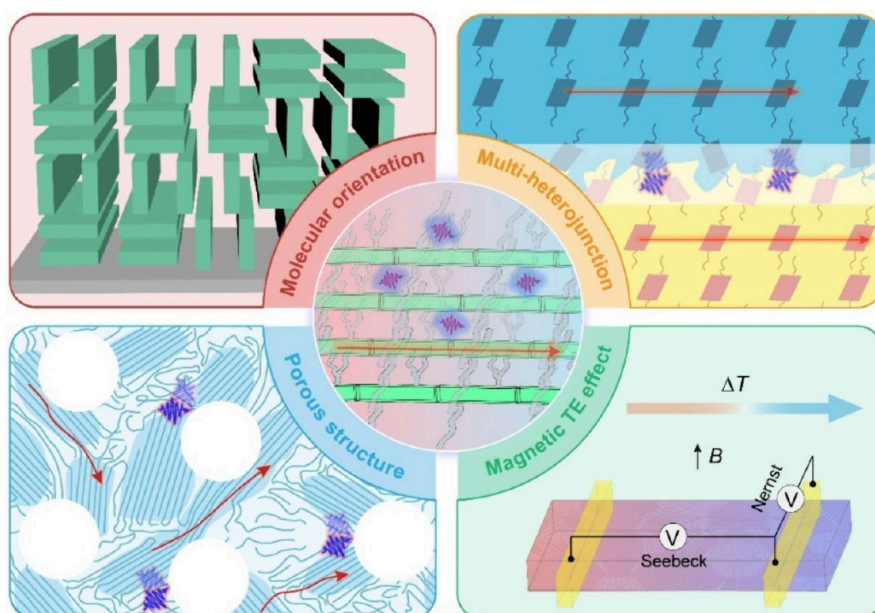


Figure 1. Molecular assembly modulation for developing high-performance and multifunctional organic thermoelectrics.

- Zhao, Y.; Li, Z. Y.; Wang, D. Y.; Zhang, X.; Ji, Z.; Niu, L. X.; Di, Y. Q.; Guan, Y.; Liu, L. Y.; Zou, Y.; et al. High Performance and Colorful Polymer Thermoelectrics with Imprinted Porous Film. *Adv. Mater.* **2024**, *36*, e2407692.³ This work highlights that polymeric thermoelectric materials with a periodic nanoscale porous structure can substantially reduce thermal conductivity without significantly compromising charge transport, thereby achieving an enhancement in thermoelectric performance.
- Ma, Y. Q.; Ren, X. L.; Zou, Y.; Zhao, W. R.; Wang, D. Y.; Ji, Z.; Fan, J. C.; Yan, C. Y.; Xiang, L. Y.; Ge, G. Y.; et al. Observation of anomalously large Nernst effects in conducting polymers. *Nat. Commun.* **2025**, *16*, 1435.⁴ This work reports anomalously large Nernst effects in doped conjugated polymers arising from quasi-1D transport and less ordered assembly, demonstrating that soft matters can govern unconventional transverse thermoelectric behavior.

1. INTRODUCTION

Harvesting low-grade heat into electricity using flexible and lightweight materials is pivotal for the advancement of wearable and portable energy technologies.⁵ Flexible thermoelectric (TE) materials have attracted growing attention, as they enable continuous energy harvesting as well as active cooling in next-generation soft electronic systems.^{6–8} Polymeric TE materials hold remarkable promise for meeting these demands, as their inherently weak intermolecular interactions endow them with exceptional mechanical flexibility, low density, and solution processability that align with the requirements of skin-integrated devices.⁹ Nevertheless, their TE performance remains significantly inferior to that of their inorganic counterparts, which poses a persistent challenge for their practical application. The performance of a TE material is commonly evaluated by the dimensionless figure of merit ZT , defined as $\sigma S^2 T / \kappa$, where S is the Seebeck coefficient, σ is the electrical conductivity, κ is the thermal conductivity, and σS^2 represents the power factor (PF).¹⁰ These parameters are intricately interdependent, as enhancing the electrical transport that governs PF often conflicts with strategies aimed at

reducing κ . Although the “phonon-glass electron-crystal” (PGEC) concept has been proposed as an ideal paradigm integrating crystalline order for efficient electronic transport with glass-like disorder for phonon suppression,¹¹ the structural demands for achieving such dual functionality are fundamentally contradictory, posing a persistent challenge for the design of high-performance TE materials.

Molecular design and chemical doping have long served as cornerstone strategies for enhancing the TE performance of polymeric materials. Through precise engineering of conjugated backbones^{12–14} and side chains,^{15–18} alongside the development of novel dopants¹⁹ and optimized doping protocols,^{20–22} it has become possible to finely tune electronic structures, boost carrier mobility and achieve higher doping level and efficiency. These advancements have progressively clarified structure–property relationships, thereby underpinning continuous improvements in the TE performance. Beyond chemical modulation, at the molecular level, the transport of charge carriers and phonons occurs within a three-dimensional material structure composed of many interacting molecules. As a result, controllable intermolecular transport is crucial, positioning the molecular assembly as an equally important strategy that critically influences TE performance. Recent studies have highlighted the decisive roles of solution-state preaggregation,^{23,24} supramolecular interactions,^{25,26} and multiscale ordering²⁷ in shaping transport-relevant domains. Strategies such as solvent-programmed crystallization,²⁸ confinement-induced alignment,²⁹ epitaxial growth,³⁰ and templated nucleation³¹ enable the formation of extended π -stacked networks with enhanced coherence length and reduced paracrystallinity. By tailoring these structural motifs, molecular assembly can affect charge-transport pathways and modulate the propagation of thermal vibrations, thereby influencing the key parameters governing TE conversion. Although the importance of molecular assembly is conceptually well-recognized, systematic investigations remain limited, leaving substantial opportunities to exploit assembly controlled structural evolution for further breakthroughs in TE performance.

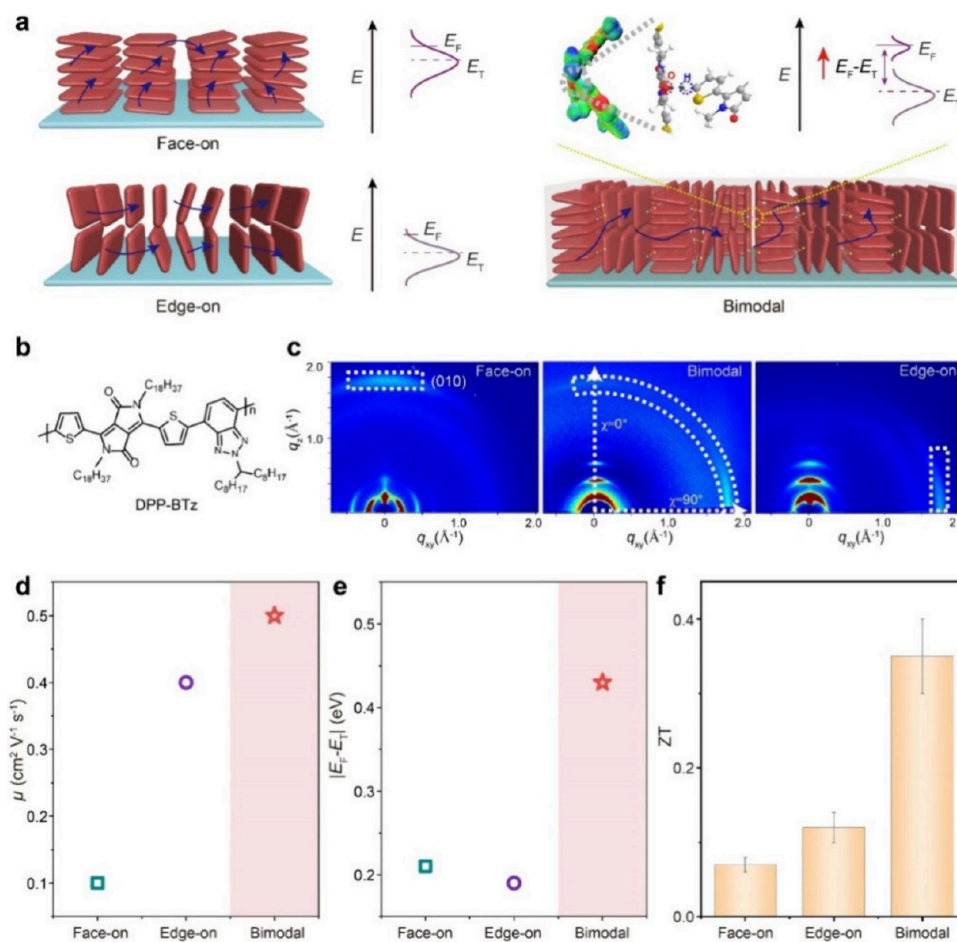


Figure 2. (a) Schematic of molecular-orientation engineering. (b) Chemical structure of DPP-BTz. (c) GIWAXS 2D diffraction patterns of DPP-BTz thin films exhibiting face-on, bimodal, and edge-on orientations. (d) Charge-carrier mobilities and (e) theoretically calculated variations in $|E_F - E_T|$ for DPP-BTz films with face-on, edge-on, and bimodal orientations. (f) Corresponding ZT values of DPP-BTz films with distinct molecular orientations. Data in panel (f) are presented as the mean $\pm$ standard deviation from three independent devices. Reproduced with permission from ref 2. Copyright 2023 Wiley-VCH.

In this Account, we highlight our recent advances in modulating molecular assembly to develop high-performance polymeric TE materials along with our efforts to uncover new physical functionalities in polymer thermoelectrics (Figure 1). To overcome the intrinsic S - σ trade-off, we developed a mixed molecular-orientation strategy in which weak interfacial hydrogen bonding synergistically enhances charge-carrier mobility, optimizes doping efficiency, and modulates the electronic density of states (DOS). This cooperative effect enables simultaneous improvements in both S and σ . We further introduced a polymeric multi-heterojunction (PMHJ) architecture to regulate thermal transport by creating nanoscale interfaces that intensify phonon scattering through an interface-diffusion-enhanced size effect. This approach substantially suppresses κ_L and yields a record-high ZT of 1.28, comparable to that of state-of-the-art inorganic thermoelectrics at the near room temperature region. To efficiently construct micro/nanostructures capable of tuning both electronic and thermal transport in polymer systems, we employed nano-imprinting to fabricate periodic porous polymer films. The resulting phonon-boundary scattering significantly reduced κ , and this approach exhibited broad applicability across different polymer systems. Finally, by leveraging the one-dimensional transport characteristics and inherently less-ordered molecular assembly of polymers, we observed anomalously large Nernst

coefficients, 2 to 3 orders of magnitude higher than classical Fermi-liquid predictions, unveiling new opportunities for advancing transverse TE effects in organic materials.

2. RELATIONSHIP BETWEEN TE PARAMETERS AND AGGREGATE STRUCTURE

Molecular assembly arises from the intricate interplay among molecule–molecule, molecule–substrate, and molecule–solvent interactions,³² and these multiscale interactions are tightly coupled to the key transport parameters (σ , S , κ) of polymeric TE materials. The electrical conductivity σ follows $\sigma = \mu n q$, where μ , n , and q denote the carrier mobility, carrier concentration, and elementary charge, respectively. Molecular orientation and alignment directly regulate μ ,^{33,34} while molecular assembly also influences dopant–polymer interactions,³⁵ thereby modulating doping efficiency and tuning n . Meanwhile, S depends on both n and the distribution of the electronic DOS. Beyond its influence on the doping level and thus n , molecular assembly reshapes the degree of structural disorder and the local energetic landscape. Such changes modulate the energetic disorder and the energy dependence of the DOS, offering additional pathways for tailoring S . The thermal conductivity κ consists of electronic (κ_e) and lattice (κ_L) contributions. κ_e scales with σ , whereas κ_L is governed by

phonon mean free paths, phonon velocities, and related parameters.³⁶ Multilevel structural features arising from molecular assembly, including grain boundaries, chain folds, twisted segments, and phase-separated interfaces, serve as effective phonon-scattering centers, substantially affecting phonon mean free paths and thereby κ_L .³⁷

In principle, the PGEC paradigm provides an ideal, conceptual framework for TE materials; however, its direct realization in conjugated polymers is intrinsically limited. Owing to weak intermolecular interactions and the absence of long-range periodicity, charge transport in polymers is governed by a combination of efficient intrachain transport and interchain hopping rather than a true electron-crystal state. Accordingly, polymeric systems can exhibit efficient charge transport along preferential pathways, which are enabled by backbone conformational regulation and chain alignment. Meanwhile, the remaining structural heterogeneities, manifested as conformational fluctuations, interfacial mismatch, and compositional contrast, serve as effective phonon-scattering centers and gives rise to intrinsically low κ . Building on these insights, a variety of molecular-assembly directed approaches have been developed to tailor charge transport in conjugated polymers. Early efforts were largely aimed at improving polymer chain crystallinity, which strongly affects μ . High-temperature rubbing, for example, was shown by Brinkmann and co-workers to generate highly oriented P3HT films with improved σ ,³⁸ while Müller et al. demonstrated that tensile drawing can induce uniaxial alignment and enhance TE performance along the stretching direction.³⁹ Complementing these physical methods, Chen et al. introduced an organic small-molecule epitaxy strategy, employing TCB as a crystallization template to produce highly oriented polymer films with significantly improved transport characteristics.³⁰ These advances collectively highlight the utility of controlled molecular assembly in optimizing chain conformation, enhancing interchain interactions, and promoting favorable mesoscale organization. However, the transport parameters that govern TE performance are inherently coupled rather than independent, and improving one parameter in isolation rarely produces transformative enhancements. Therefore, realizing high-performance polymeric thermoelectrics through molecular-assembly engineering requires more sophisticated and integrated control over the multilevel structural organization.

3. MIXED MOLECULAR ORIENTATION TO OVERCOME THE S - σ TRADE-OFF

Molecular orientation, defined as the relative spatial arrangement between the conjugated backbone and the substrate, has been extensively studied due to its significant impact on charge-transport behavior.^{33,34} The two typical predominant molecular orientation modes are face-on and edge-on. In conventional applications such as organic photovoltaics^{40,41} and organic field-effect transistors,^{42,43} films exhibiting a single dominant orientation have long been adopted to enhance device performance. However, for TE applications, a single molecular orientation only provides restricted charge-transport pathways and a specified sole electronic structure, making it difficult to modulate the TE characteristics by simultaneously optimizing μ and electronic DOS. This limitation ultimately constrains the enhancement of PF. To overcome these challenges, we introduce the concept of bimodal orientation, which refers to the coexistence of both face-on and edge-on

molecular orientations² (Figure 2a). Bimodal orientation assemblies formed through weak intermolecular interactions can create a three-dimensional, interpenetrating charge-transport network that provides efficient hopping bridges for charge carriers and significantly enhances μ . In addition, bimodal orientation can increase the energetic disorder and modulate the electronic DOS, thereby boosting S and enabling a substantial improvement in overall TE performance.

Translating this concept into practice requires precise control over the intermolecular interactions. We selected a diketopyrrolopyrrole (DPP)-based polymeric semiconductor, DPP-BTz (Figure 2b), as a model system because its molecular orientation can be tuned from face-on to edge-on through dynamic templating combined with a solution shearing method.⁴⁴ Under conventional solution shearing on an octadecyltrichlorosilane (OTS)-treated substrate, DPP-BTz naturally adopts a predominantly face-on orientation. The introduction of a dynamic template above the substrate, however, provides an additional means to regulate the interplay between polymer chains and the templating layer while simultaneously tailoring the shear fields experienced during film formation. This combined modulation establishes a precise route to manipulate intermolecular interactions and enables reversible adjustment of the molecular orientation from face-on to edge-on. In the dynamic templating process, the polymer solution is deposited onto an ion-gel substrate composed of EMIM:TFSI and PVDF, followed by solution shearing. The two components interact differently with the polymer, with EMIM:TFSI engaging more strongly with the conjugated backbone of DPP-BTz and PVDF interacting preferentially with its side chains. By varying the EMIM:TFSI to PVDF ratio in the ion gel, the intermolecular interactions can be finely modulated, enabling the formation of films with different molecular orientations (Figure 2c). According to the grazing incidence wide-angle X-ray scattering (GIWAXS), the Herman's orientation from the (010) peak increased from 0.38 to 0.48 with increasing ionic liquid content, indicating significant molecular orientation variation. When the ionic liquid content is adjusted to 50 wt %, a bimodal orientation emerges, producing a film that contains approximately 37% face-on and 63% edge-on orientations.

The resulting bimodal-oriented films exhibited higher μ and enhanced S under identical doping conditions than their single-orientation counterparts. Microstructural and theoretical analyses revealed the formation of weak C—H...O hydrogen bonds at the interfaces between differently oriented domains. These interfacial interactions expanded the lamellar spacing, facilitated diffusion of dopant, and improved doping efficiency. Moreover, the orientation-dependent electronic structures induced an energy-level splitting between Fermi level (E_F) and transport level (E_T), enabling decoupled and simultaneous enhancements in S and σ (Figure 2d,e). As a result, the bimodal-oriented DPP-BTz films achieved a maximum PF of $346 \mu W m^{-1} K^{-2}$ and a peak ZT of 0.40, more than four times higher than that of single-orientation films, demonstrating the critical role of molecular orientation in optimizing polymeric TE performance (Figure 2f).

Despite these advances, establishing reproducible and predictive control over the molecular orientation remains highly challenging. Studies by Pei and Neher have indicated that polymer orientation is strongly dictated by the preaggregation behavior of conjugated chains in solution, which is highly sensitive to solute–solvent interactions^{23,24}

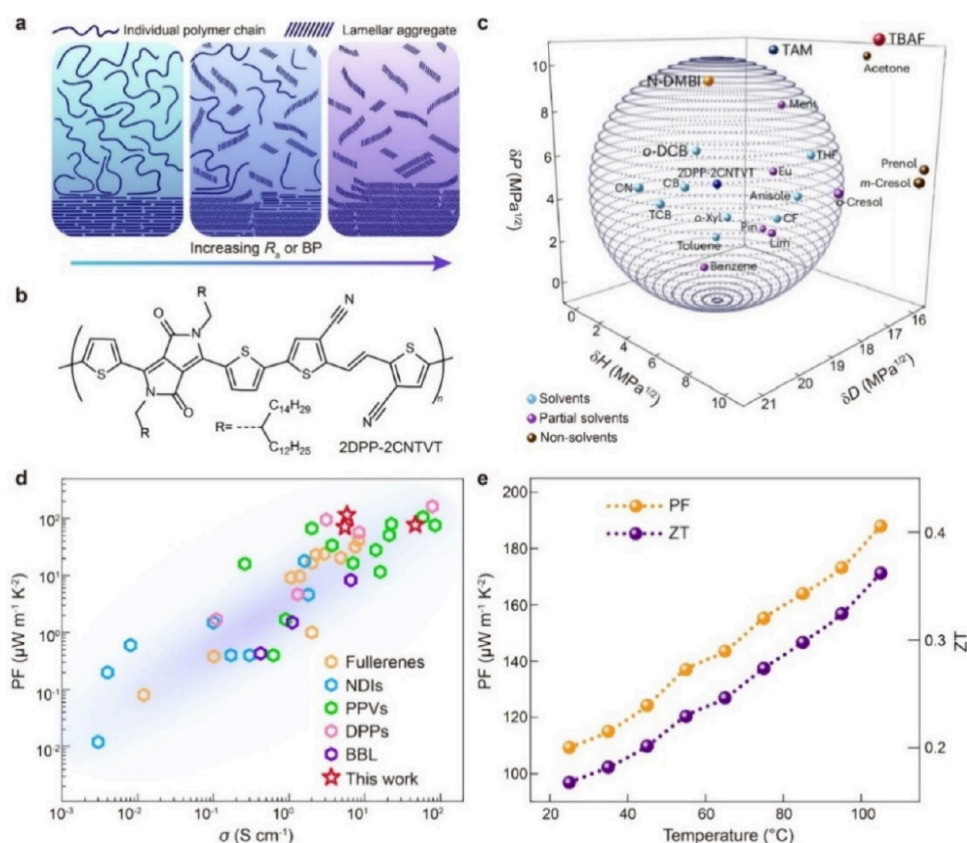


Figure 3. (a) Illustration of the molecular force-induced anisotropic assembly, showing how solution aggregates affect molecular orientation into the solid film. (b) Chemical structure of 2DPP-2CNTVT. (c) Hansen-type solubility sphere of 2DPP-2CNTVT measured at ambient temperature. (d) Comparison of PF and σ among state-of-the-art n-type OTE materials. (e) Evolution of PF and ZT with temperature for the 2DPP-2CNTVT thin film. Reproduced with permission from ref 45. Copyright 2025 Springer-Nature.

(Figure 3a). The Hansen solubility parameters provide a quantitative framework to describe these interactions, where the Hansen distance (R_a) between solute and solvent centroids governs polymer–solvent miscibility and preaggregation, ultimately determining the resulting film orientation. Building on this concept, Rosas Villalva and colleagues⁴⁵ selected 2DPP-2CNTVT as a representative high-performance TE polymer (Figure 3b) and investigated its assembly behavior. They developed a molecular-force-driven anisotropic self-assembly strategy that integrates Hansen solubility parameters with solvent boiling points to predict and control polymer orientation (Figure 3c). This approach enables the selective formation of films with adjustable fractions of face-on and edge-on domains through intermolecular force-driven anisotropy without requiring any post-treatment. It thereby enhances the in-plane delocalization length. Notably, such synergistic control effectively mitigates the conventional trade-off relationship between S and σ , achieving a maximum PF of $115 \mu W m^{-1} K^{-2}$ and a ZT of 0.17 at room temperature for the doped n-type 2DPP-2CNTVT:N-DMBI system (Figure 3d,e).

Although significant progress has been made, current molecular-orientation strategies have yet to fully exploit their potential for use in polymeric TE materials. Most efforts remain limited to a narrow set of polymers with favorable solubility and processability, leaving orientation control in the most promising high-performance systems largely unexplored. Expanding this scope will require general and robust design rules, such as embedding multiple weak and programmable interactions during synthesis or assembly, to achieve reliable

orientation tuning across diverse polymer structures. Meanwhile, the mechanisms connecting the molecular orientation to TE transport remain poorly understood. Critical questions include how interfaces between different orientations modulate the electronic DOS and redirect charge pathways and how nanoscale orientation-dependent domains collectively shape macroscopic transport. Addressing these challenges and precisely manipulating such effects will be essential for future breakthroughs. Ultimately, advancing orientation engineering will depend on enhanced controllability, broader applicability to high-performance polymers, and deeper mechanistic insight, offering pathways not only to understand molecular assembly transport relationships but also to unlock novel transport regimes in next-generation organic TE materials.

4. SUPERLATTICE-LIKE ASSEMBLY FOR LEAPED TE CONVERSION

Molecular orientation engineering has demonstrated a remarkable ability to overcome the S - σ trade-off and enhance PF, while further progress relies on precise control of κ . Polymeric TE materials are often regarded as intrinsically poor thermal conductors due to weak intermolecular interactions, which has led to the assumption that fine-tuning κ is unnecessary. In addition, reliable measurement of thermal conductivity in polymer thin films remains challenging due to complex device architectures and inherent uncertainties in experimental protocols.⁴⁶ As a result, research has largely focused on optimizing molecular order to improve electronic transport, leaving the fundamental aspects of thermal transport

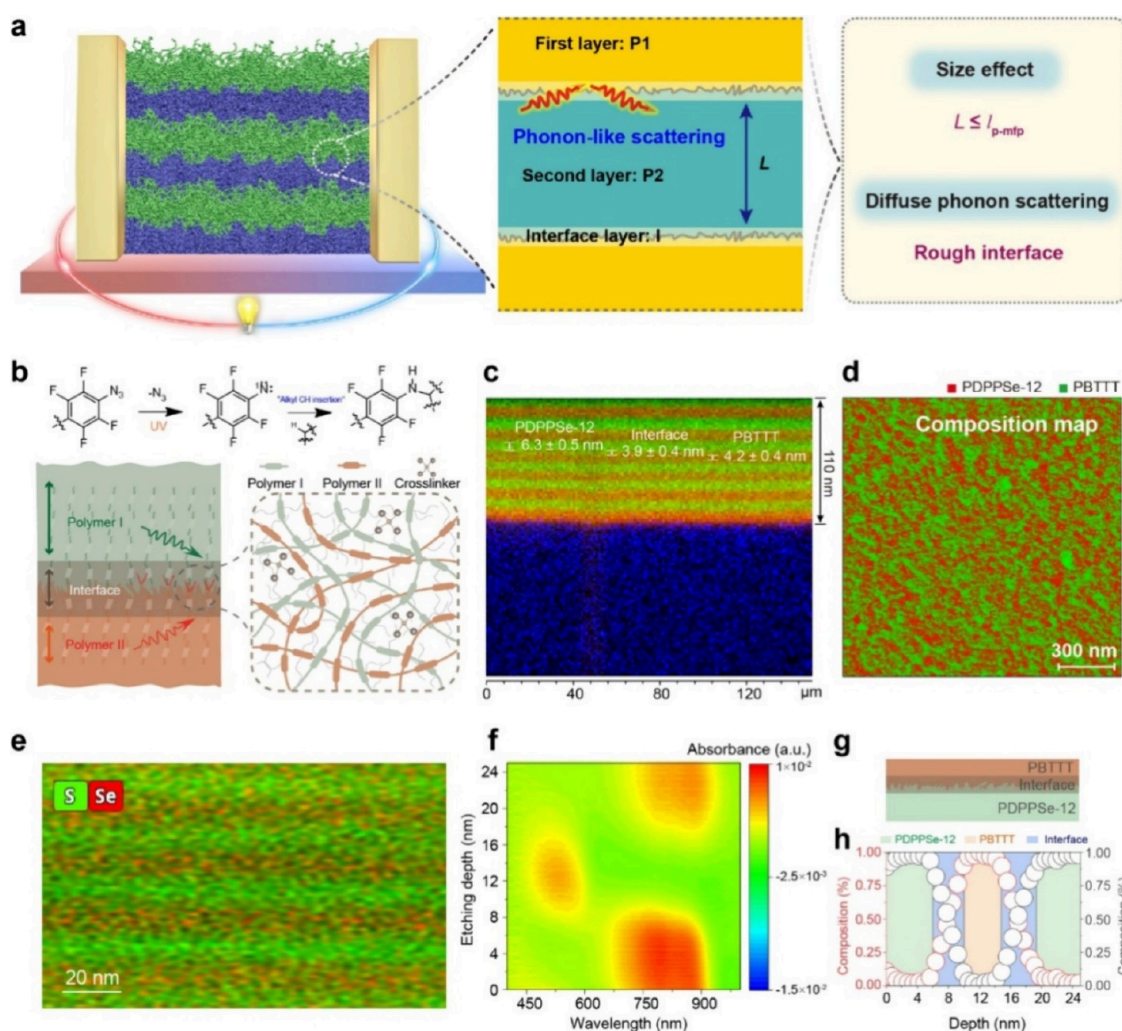


Figure 4. (a) Schematic of the PMHJ architecture and the key mechanisms by which it suppresses κ_L . (b) Illustration of the photo-cross-linking approach for constructing PMHJ structures and the resulting interpenetrated interfaces. (c) TOF-SIMS cross-sectional images of the PDPPSe-12:PBTTT PMHJ. (d) Photoinduced force microscopy (PiFM) compositional maps of PDPPSe-12 and PBTTT across the PMHJ interface. (e) EDS mapping results of the distribution of S and Se elements. S and Se elements were utilized as labels for PBTTT and PDPPSe-12, respectively. (f) Optical intensity contours reconstructed from absorption spectra of PMHJ films. (g) Interface schematic of PMHJ films. (h) Variation of the two-polymer composition at different etching depths. Reproduced with permission from ref 1. Copyright 2024 Springer-Nature.

largely unexplored. In contrast, inorganic thermoelectrics have long harnessed a range of nanoscale structural strategies, deliberately introducing interfaces, defects, and dislocations to scatter phonons without compromising charge transport, achieving substantial enhancements in TE performance.^{47,48} Translating such approaches to organic systems, however, is inherently challenging, as the weak van der Waals interactions and limited structural rigidity of polymeric materials make the controlled introduction of effective phonon-scattering centers far more difficult, thus presenting a major obstacle to the rational design of high-performance organic TE materials.

Inspired by the design principles of inorganic superlattices, we developed a PMHJ architecture featuring periodic dual-heterojunctions, where each period consists of two distinct polymer layers separated by interpenetrating interfaces¹ (Figure 4a). The polymer layers facilitate dual-channel charge transport, while the interfaces enhance phonon scattering, establishing a delicate balance between electronic and thermal transport. In particular, the κ_L can be effectively reduced through size-effect-enhanced interfacial scattering when the layer thickness approaches the phonon mean free path along

the conjugated backbone and π - π stacking directions. This structural combination offers a powerful approach to decoupling charge and phonon transport, ultimately optimizing TE performance.

To realize the PMHJ concept, we employed photo-cross-linking together with two representative polymeric TE materials, PDPPSe-12 and PBTTT, to achieve precise periodic assembly (Figure 4b). The resulting films form hierarchical heterostructures while retaining bulk-like heterojunction characteristics at the interfaces (Figure 4c,d). When the layer thicknesses are close to the phonon mean free path, specifically 6 nm for PDPPSe-12 and 4 nm for PBTTT, interfacial scattering becomes significantly enhanced and reduces κ_L by more than 70% to $0.10 \text{ W m}^{-1} \text{ K}^{-1}$ (Figure 5a,c). The in-plane κ was measured using both the 3ω method with a Linseis TFA system and suspended microdevices based on Fourier's law, and the use of substrate and reference calibrations provides cross-validated and reliable values. The Lorentz factor in the Wiedemann–Franz law remains debated, and separating electronic and lattice contributions can be ambiguous.⁴⁹ The thermal conductivity of undoped films was used to estimate the

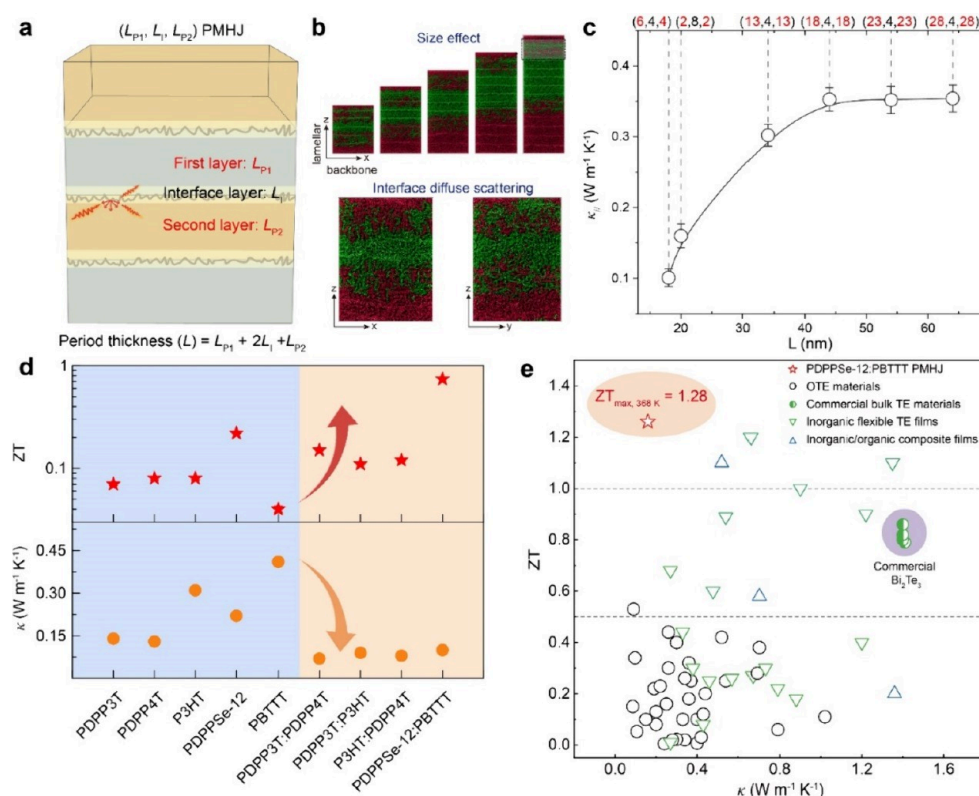


Figure 5. (a) Schematic illustration of the periodic PMHJ architecture. (b) Simulated structures used to investigate size effects and interfacial diffuse scattering within the PMHJ architecture. (c) Evolution of κ_L as a function of periodic thickness. Data in panel (c) are presented as mean $\pm$ standard deviation from three independent devices. (d) General applicability of the PMHJ architecture for reducing κ_L and enhancing ZT. (e) Comparative plot of κ and ZT for room-temperature organic TE materials, commercial inorganic TE materials, inorganic plastic TEs, and organic-inorganic hybrid TE materials. The CYTOP-M (3:1)-encapsulated PMHJ film exhibited negligible variation in TE performance after a 10 h duration in vacuum conditions. Reproduced with permission from ref 1. Copyright 2024 Springer-Nature.

lattice component and determine the electronic contribution in the doped films. Molecular dynamics simulations show that size effect and interfacial diffuse scattering contribute approximately 34% and 66% of the reduction in κ_L (Figure 5b). This strategy is broadly applicable across different polymer combinations and consistently produces reductions in κ_L of 36–76% together with 2- to 3-fold improvements in ZT (Figure 5d). The PMHJ films based on PDPPSe-12 and PBTTT achieve a maximum PF of 628 $\mu\text{W m}^{-1} \text{K}^{-2}$, and the configuration with layer thicknesses of 6, 4, and 4 nm reaches a peak ZT of 1.28, which is comparable to commercial inorganic TE materials operating at near room temperature region (Figure 5e). It should be noted that no widely accepted thermal transport model currently exists for conjugated polymers. Drawing conceptual insight from the Allen–Feldman framework for amorphous solids,⁵⁰ heat conduction in conjugated polymers may involve multiple vibrational contributions. Phonon-like transport is expected along rigid conjugated backbones, whereas diffusive vibrations are more likely associated with π – π -stacked domains and side chains. However, quantitative separation of propagating and diffusive contributions remains an open challenge.

Although PMHJ architecture has already demonstrated remarkable TE performance, considerable potential remains for further improvement through multi-nanoscale hierarchical molecular assembly. In this architecture, the selection of polymer constituents and cross-linking agents demands meticulous refinement, as parameters including energy-level alignment, structural heterogeneity, and interfacial morphology

collectively exert profound influence on both charge transport and phonon scattering. Moreover, in PMHJ architecture, phonon scattering is dominated by size effects and interfacial diffuse scattering, motivating the use of vertical hierarchical configurations where higher interface density can further suppress κ . However, preserving σ then hinges on precisely controlling the vertically assembled nanostructure to sustain efficient charge transport. Equally critical is the development of fundamental insight into how buried interfaces orchestrate carrier dynamics and phonon propagation on the molecular scale. Addressing these questions will be essential for the rational design of polymeric TE materials capable of achieving a more transformative performance.

5. TOWARD EASY-TO-PROCESS POROUS FILMS

A key factor in advancing polymeric thermoelectric materials toward practical applications is the development of simple, reliable, and easily scalable processing methods. Although the PMHJ architecture has delivered impressive TE performance, its fabrication often requires hundreds of deposition cycles, which underscores the need for simpler and more scalable approaches to manipulate polymer micro- and nanostructures. In the framework of the phonon-glass electron-crystal model, ideal thermoelectric materials should combine amorphous-level phonon scattering with crystalline-level charge transport. Porous polymer films naturally meet this requirement because the numerous interfaces formed between the polymer matrix and pores can strongly enhance phonon-boundary scattering.

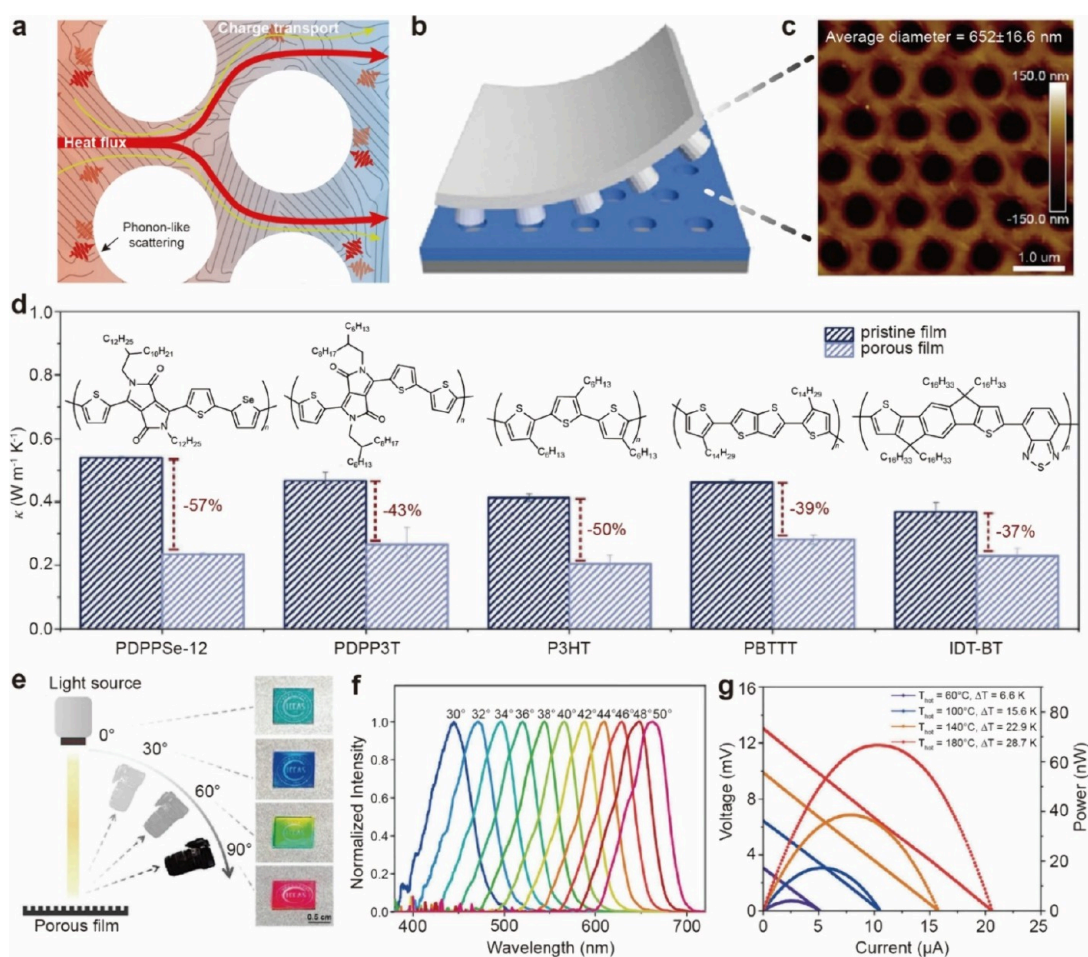


Figure 6. (a) Schematic illustration of a porous polymeric TE film, where the porous structure induced phonon-like scattering effectively suppresses κ_L . (b) Schematic of the nanoimprinting process for fabricating porous polymer films. (c) AFM image of the nanoimprinted porous film with 652 ± 16.6 nm pores. (d) General applicability of the porous structure for lowering κ across multiple polymers; molecular structures of five polymers are shown in the inset. Data in panel (d) were obtained from at least three independent samples and are presented as mean $\pm$ standard deviation. (e) Schematic and optical images showing angle-dependent structural coloration generated by the porous architecture. (f) Spectral response of the structural color at viewing angles from 30° to 50° . (g) Output voltage and power of the integrated TE generator under different temperature gradients. The flexible device was continuously operated on a hot plate at 100°C in air for 12 000 s during the measurement. Reproduced with permission from ref 3. Copyright 2024 Wiley-VCH.

Building on this principle, we employed nanoimprinting to create periodic porous polymer films³ (Figure 6a–c). Compared with dense films, the periodic porous architecture markedly reduced the κ of PDPPSe-12, with larger pore diameters enhancing phonon scattering and further lowering κ . Meanwhile, efficient charge transport was preserved, yielding a substantial improvement in the ZT. The PDPPSe-12 film with a $1\ \mu\text{m}$ period, 652 nm pore diameter, and 25.5% porosity exhibited a κ of $0.23\ \text{W m}^{-1} \text{K}^{-1}$, which is over 50% lower than that of pristine film, and achieved a peak ZT of 0.52 at 363 K, representing a 1.7-fold enhancement. The reduction of κ was observed in several other organic TE systems with porous structure (Figure 6d). Accounting for possible measurement uncertainties arising from the device vibration and systematic errors associated with different characterization techniques, the in-plane κ of these films was probed using Linseis TFA and transient infrared measurements, both revealing a pronounced suppression of thermal transport. Furthermore, the periodic porous architecture fulfilled the grating diffraction criteria, yielding vivid structural colors across the visible spectrum with pronounced angle-dependent optical responses. Specifically,

with a grating period of 866 nm with perpendicular incident light, the film's color transitions gradually from blue to red as the observation angle ranges from 30° to 50° (Figure 6e,f). When integrated on human skin, a device comprising 16 p-type legs delivered a maximum output power of 67 nW (Figure 6g), demonstrating the strong potential of this porous architecture for high-performance wearable and skin-conformal energy-harvesting systems.

These observations underscore the considerable potential of porous architectures for polymeric thermoelectrics while simultaneously highlighting the fundamental challenges posed by their inherently dualistic nature. Although introducing abundant pores remains an effective strategy for reducing κ , the resulting increase in interfacial density hinders charge transport, thereby limiting improvements in ZT. Looking ahead, strengthening the tightness and coherence of molecular packing will be essential for counteracting the reduced electronic pathways imposed by porous frameworks, and controllable nanophase separation during pore formation offers a promising route to tune local environments, reinforce backbone planarity, and guide chain packing, potentially

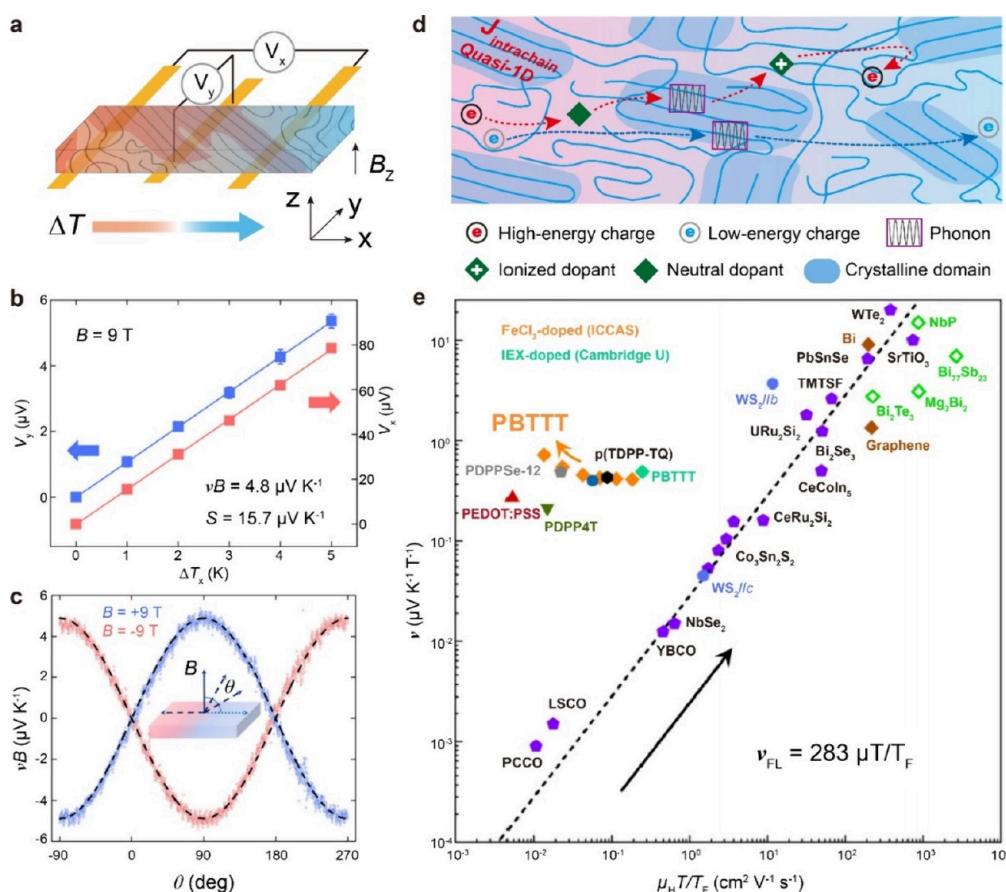


Figure 7. (a) Schematic illustration of the OTE Nernst device. Electrical signals were applied using an Agilent B2902A and measured with a Keithley 2182A nanovoltmeter with low-noise wiring limiting the system noise below 10 nV and a typical Nernst measurement taking about 1 h at a sweep rate of 200 Oe s^{-1} . For polymer Nernst effect measurements at 300 K, the temperature gradient was carefully maintained below 10 K to prevent significant dedoping.⁴ (b) Temperature-dependent Seebeck (red) and Nernst (blue) voltage responses under varied thermal gradients. (c) Angular dependence of the Nernst coefficient ν , plotted as a function of the angle θ between the rotating temperature gradient and the out-of-plane magnetic field. The error bars in ν were determined from the uncertainty in the extraction of the electromotive force and represent one standard deviation. (d) Conceptual illustration highlighting the broad and strongly energy-dependent quasiparticle scattering mechanisms present in heavily doped polymeric TE films. (e) Relationship between ν and $\mu_H T/T_F$ —where μ_H denotes the Hall mobility, T is absolute temperature, and T_F is the Fermi temperature—compiled for multiple polymers, bismuth, graphene, and additional materials reported in the literature. Reproduced from ref 4. Available under a CC BY-NC-ND 4.0 license. Copyright 2025 Springer Nature. Yingqiao Ma et al.

enhancing μ even within porous matrices. At the same time, porous polymers provide a versatile platform for orchestrating multiscale phonon-scattering mechanisms, where interface-driven scattering shaped by pore geometry and spatial order, together with strong size-effect scattering in nanothroat regions, opens broad opportunities for suppressing κ . Local confinement may further modulate packing to intensify phonon–phonon scattering, adding another lever for thermal regulation. The interplay between constrained electronic pathways and tunable phonon scattering thus defines a forward-looking design space in which precisely engineered porous architectures may unlock the high performance polymeric thermoelectrics.

6. UNVEILING MAGNETIC TE EFFECTS IN POLYMERIC AGGREGATES

While molecular assembly has been proven to be essential for enhancing TE performance, its influence also extends to the emergence of unconventional physical responses in polymers. In particular, such an assembly can give rise to behaviors that are inaccessible in more rigid inorganic systems. Among these, magnetic TE signals such as the Nernst effect provide a

powerful probe of charge and heat transport and offer opportunities for expanding the functionality of flexible thermoelectrics. Landau's Fermi liquid (FL) theory predicts an almost negligible Nernst coefficient (ν) in low-mobility and weakly ordered systems.⁵¹ This expectation, supported by the well-ordered crystal lattices and high mobilities of inorganic TE materials where Nernst signals are commonly observed, has long discouraged exploration of magnetic TE behavior in polymeric systems. However, the situation in conjugated polymers is fundamentally different. Their covalently bonded backbones resemble those of inorganic semiconductors, but the presence of multiple weak intermolecular interactions drives a hierarchical assembly that produces alternating ordered and disordered domains. In principle, this assembly defined morphology results in anisotropic, quasi-one-dimensional transport, where rapid intrachain conduction and much slower interchain hopping deviate strongly from FL behavior and instead resemble Luttinger-liquid (LL) models.⁵² Owing to confinement in the other two dimensions, such systems are expected to exhibit enhanced scattering and strong quasiparticle interactions,^{53,54} which can theoretically give rise to anomalously large Nernst responses, as ν reflects carrier

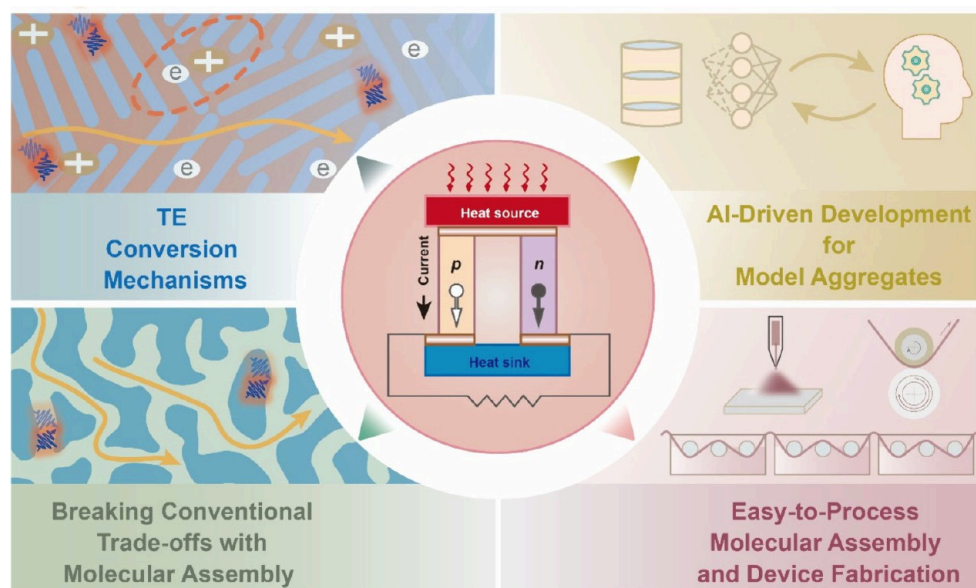


Figure 8. Key aspects for advancing organic thermoelectrics.

transport entropy and has been reported in various one-dimensional systems.^{55–57} Such assembly driven transport anisotropy suggests that polymers can exhibit a nontrivial Nernst effect despite their modest mobility, making it both plausible and highly informative as an intrinsic consequence of molecular assembly.

Translating this potential into measurable signals, however, is challenging because localized charge carriers tend to screen transverse voltages, resulting in intrinsically low signal-to-noise ratios. To overcome this limitation, we performed repeated field-sweeping measurements under strong superconducting magnetic fields (± 9 T) to enhance the signal-to-noise ratio⁴ (Figure 7a). Remarkably, a pronounced anomalous Nernst effect was observed with $\nu = 0.53 \mu\text{V K}^{-1} \text{T}^{-1}$ and $S = 15.7 \mu\text{V K}^{-1}$, yielding a $\nu B/S$ ratio of 30.6% (Figure 7b). Angular-dependent measurements confirmed a sinusoidal dependence of ν on the relative orientation between the magnetic field and temperature gradient, verifying its intrinsic magnetic origin (Figure 7c). The unusually large Nernst response—2–3 orders of magnitude greater than FL predictions—likely arises from the quasi-1D transport nature of doped polymers, where strong energy-dependent scattering and quasiparticle interactions enhance carrier entropy and transverse TE coupling (Figure 7d). This observation of an anomalously large Nernst effect in conducting polymers, reproducible across multiple molecular systems and microstructures (Figure 7e), unveils a fundamentally distinct transport regime and opens new possibilities for the design of transverse organic TE devices.

Looking ahead, the discovery of the anomalous Nernst effect in conjugated polymers opens a window into the broader landscape of spin, heat, and charge coupling in soft materials. A key challenge lies in the weak magnetic interactions of polymers, which limit spin polarization and the detection of signals. Overcoming this challenge requires introducing controlled spin moments without disrupting electronic transport. This can be achieved, for example, by incorporating trace magnetic elements that enhance spin density and spin-charge coupling, thereby amplifying transverse TE responses. In parallel, a deeper understanding of the entropy flow associated with charge transport, particularly the energy distribution and

transport entropy of carriers, will be essential for developing predictive models of transverse thermopower in quasi-1D polymers. This emerging direction may enable spin-polymer TE devices capable of magnetic-field sensing, directional heat routing, and novel low-power spintronic functions, thereby opening a new frontier in organic TE research.

7. CONCLUSION AND OUTLOOK

Guided by fundamental chemical principles, we have developed novel concepts for fine-tuning molecular orientation and designing periodic nanostructures to leapfrog TE performance and unlock new functionalities in polymeric semiconductors. By virtue of our effects and systematic study by other groups, the ZT values have already surpassed 1.0. Despite these promising advances, several fundamental challenges remain. What molecular structure and assembly architectures can ideally fulfill the PGEC paradigm—achieving the dual aims of low thermal conductivity and high conductivity? How can we reliably describe and control thermal transport in polycrystalline polymer films where grain boundaries and structural heterogeneity critically influence performance? Beyond conventional power generation, what distinctive application scenarios can organic TE materials serve? Achieving these goals will require not only the molecular design of new materials but also a deeper mechanistic understanding of the structure–property relationships that govern their performance. Addressing these open questions may pave the way toward a new era of organic thermoelectrics.

To further enhance the performance of organic TE materials and realize practical applications, further efforts in the near future should be devoted to the following aspects (Figure 8).

7.1. Elucidating TE Conversion Mechanisms in Polymers

A deep mechanistic understanding of the key processes governing TE conversion in polymers is vital to achieving target performance. This requires unraveling the interplay between charge carrier transport, thermal vibration scattering, and charge-phonon coupling, all of which collectively determine the ZT . While charge transport in polymers has been extensively investigated, thermal transport remains a

critical yet poorly explored frontier. The challenge arises from the amorphous or semicrystalline nature of polymer, where structural disorder disrupts phonon propagation and conventional crystalline models fail to capture the full physics. Quantitative description of phonon and phonon-like quasi-particle scattering is still lacking. Progress will demand advanced experimental characterization coupled with multi-scale simulations to resolve heat and charge transport at the molecular scale.

7.2. AI-Driven Design of Model Aggregates to Approach TE Conversion Limit

Historically, progress in organic TE materials has relied on empirical design and assembly. The convergence of theoretical modeling, high-throughput computational screening, and machine learning is now enabling an AI-driven design paradigm. In our systems, structural descriptors such as polymer chain orientation distribution, interfacial density, and heterojunction length scale play key roles in governing charge transport and phonon scattering. Incorporating these descriptors into AI models, it is possible to predict trends in TE performance for candidate molecules and their assembled aggregates prior to synthesis. However, AI methods currently face significant challenges due to the intrinsic structural complexity and limited data availability in conjugated polymers. High-throughput synthesis and automated measurement pipelines are therefore essential to generate reproducible data sets, enabling iterative refinement of AI predictions. Through such closed-loop workflows, it becomes possible to identify optimal polymer systems whose model aggregates approach the theoretical limits of TE conversion.

7.3. Precise Molecular Assembly to Overcome Conventional Trade-Offs

TE optimization is often constrained by intrinsic trade-offs—enhancing σ typically reduces the S and improving carrier mobility may increase κ . Overcoming these limitations requires precise control over molecular packing and orientation as well as well-controlled energetic disorder. Strategies such as homojunction architectures, block copolymer design, and controlled nanophase separation can produce anisotropic charge-transport pathways while selectively impeding phonon flow. Combining highly ordered polycrystalline domains with engineered disordered grain boundaries offers a promising avenue.

7.4. Developing Scalable Molecular Assembly Concept and Device Construction Technologies

Scalability remains essential for translating organic TE materials from prototypes into commercial applications. Molecular assembly and processing routes must preserve a high TE performance while enabling flexible, lightweight configurations suited to wearable electronics and distributed energy systems. Fiber-based architectures, printable TE inks, and roll-to-roll manufacturing techniques are especially promising. Efforts should emphasize morphological stability under large-scale processing, reproducible device performance, and seamless integration with existing electronic and textile manufacturing techniques.

Collectively, these research avenues present a coherent roadmap for the further development of organic thermoelectrics. By addressing scientific and engineering barriers in parallel, the field is well poised to deliver a new generation of

high-performance, flexible, and multifunctional organic TE devices in the near future.

AUTHOR INFORMATION

Corresponding Author

Chong-an Di — Beijing National Laboratory for Molecular Sciences, CAS Key Laboratory of Organic Solids, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, China; orcid.org/0000-0002-6183-1321; Email: dicha@iccas.ac.cn

Authors

Dongyang Wang — Beijing National Laboratory for Molecular Sciences, CAS Key Laboratory of Organic Solids, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, China

Daoben Zhu — Beijing National Laboratory for Molecular Sciences, CAS Key Laboratory of Organic Solids, Institute of Chemistry, Chinese Academy of Sciences, Beijing 100190, China; orcid.org/0000-0002-6354-940X

Complete contact information is available at:
<https://pubs.acs.org/10.1021/acs.accounts.5c00867>

Author Contributions

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Notes

The authors declare no competing financial interest.

Biographies

Dongyang Wang obtained her Ph.D. from the Institute of Chemistry, Chinese Academy of Sciences, in 2024 under the supervision of Prof. Chong-an Di and Prof. Daoben Zhu. She is currently conducting her postdoctoral research at the same institute. Her research focuses on the development of organic thermoelectric materials and devices.

Daoben Zhu is a professor and director of the Organic Solids Laboratory in the Institute of Chemistry, CAS. He was selected as an Academician of the CAS in 1997. His research interests include molecular materials and devices.

Chong-an Di received his Ph.D. degree in Chemistry from Institute of Chemistry, Chinese Academy of Sciences, in 2008. He was appointed to the position of associate professor in 2010 and promoted to professor in 2016 at ICCAS. His research currently focuses on the investigation of organic thin-film transistors and organic thermoelectric devices.

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